

Neural Network-Based Acoustic Detection of Red Palm Weevil Infestation in Coconut Trees*

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Abstract—The red palm weevil (*Rhynchophorus ferrugineus*) poses a serious threat to coconut palms, often causing irreversible damage before visible symptoms appear. Traditional inspection methods are labor-intensive and ineffective for early-stage detection. This study proposes a neural network-based approach for real-time detection of red palm weevil infestations using acoustic signals emitted by the larvae within coconut trees. Sound recordings collected under diverse environmental conditions are preprocessed and used to train a lightweight neural network model capable of identifying infestation patterns based on characteristic audio frequencies. The model's performance is evaluated using key metrics such as accuracy, precision, recall, and F1-score. Experimental results demonstrate that the proposed method effectively detects red palm weevil activity at an early stage, offering a scalable and low-cost solution for precision agriculture and timely pest management. This acoustic detection technique can be integrated into edge devices for continuous monitoring, enabling early intervention and reducing crop losses.

Index Terms—Red Palm Weevil, Coconut Trees, Acoustic Detection, Neural Network, Pest Monitoring, Precision Agriculture, Sound-Based Detection, Edge Computing.

I. INTRODUCTION

Agriculture functions as the fundamental pillar for numerous nations because it both sustains food supplies and generates work opportunities for billions of individuals around the world. The major threat to crop productivity emerges from pest infestations which lead to extensive losses both in agricultural production and economic value. The primary approach to pest detection involves human visual inspection together with expert analysis as well as field investigations. The methods require extensive human labor and consume much time while being at risk of human mistakes or inaccurate judgments. Monitoring pest activities across extensive agricultural fields becomes nearly impossible because of its size which results

in uncontrolled pest outbreaks. The combination of seasonal changes and environmental conditions makes it difficult to detect pests using manual methods which results in operational inefficiencies for large farming areas.

The Red Palm Weevil is an invasive beetle that severely damages coconut and other palm trees worldwide. Originating from South and Southeast Asia, it has spread through global trade to many regions. Females lay eggs in wounds on palms, and the larvae feed inside the tree, causing hidden damage that's hard to detect early. Visible symptoms like yellowing fronds and tree death appear only in advanced stages. Because the weevil lives mostly inside the trunk, early detection is difficult, often leading to irreversible damage that harms both agriculture and rural economies. Current detection methods, such as visual checks and traps, are limited, highlighting the need for new automated, non-invasive techniques. Researchers are exploring bioacoustic signals from larvae feeding inside trees to improve early detection and protect affected palms. The ability to detect pests early stops them from causing extensive harm while decreasing the use of chemical pesticides and enabling farmers to choose particular pest control approaches. Early detection of infestations at their starting point will lead to massive reductions in both crop health damage and financial costs. The existence of automation alongside intelligent detection systems has evolved from being helpful to becoming indispensable. Neural Networks together with IoT and Real-Time Monitoring provide automated pest detection for the 1.4. modern era. The last years have seen artificial intelligence progress through deep learning technology to develop pest detection systems based on neural networks. The precision of Convolutional Neural Networks (CNNs) allows them to detect pests accurately while determining their exact location through image analysis at high resolution. The AI models have the ability to learn without stopping which enables them to adjust to new pest species and environmental modifications

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thus achieving optimal performance in agricultural settings that constantly change. The combination of neural networks with IoT-based sensor systems and mobile applications allows for real-time monitoring and instant notifications. The combined system gives farmers valuable information that leads to quicker and more accurate intervention methods. The data collection by smart cameras and environmental sensors operates continuously while neural networks process the information to identify abnormal conditions and detect pests within the environment.

II. RELATED WORKS

The research by Soroker et al. [1] in 2010 introduced the use of acoustic sensors for detecting early-stage red palm weevil (RPW) larvae within palm trees. The team employed trunk-embedded microphones to capture larval feeding sounds, enabling effective non-invasive pest detection and early intervention before visible symptoms appeared. In 2012, Faleiro et al. [2] evaluated acoustic devices in actual coconut plantations, demonstrating that larval movement and chewing generated distinct sound signatures that differed significantly from environmental noise, such as wind or birds. Their approach illustrated how farmers could monitor infestations without causing harm to trees. Potamitis and Rigakis [3] developed in 2013 a solar-powered, autonomous sound recording system capable of real-time detection and analysis of larval sounds using Fourier and time-frequency analysis. This system enabled continuous monitoring and eliminated the need for manual inspection. Al-Manie et al. [4] in 2015 introduced hardware enhancements using directional audio sensors and amplification circuits targeting 1–5 kHz vibrations to improve detection in noisy environments. Rach et al. [5] in 2016 employed time-domain analysis and Fast Fourier Transform (FFT) for classifying RPW audio signals, resulting in improved precision and reduced false alarms. In 2017, Kumbhar et al. [6] applied Mel-frequency cepstral coefficients (MFCC) and Gaussian mixture models (GMM) for efficient sound classification, enabling lightweight deployment on embedded systems. The 2018 work by Potamitis et al. [7] integrated supervised machine learning directly into detection hardware, utilizing labeled audio clips to train classifiers and enhance real-time reliability. In 2019, Liu et al. [8] introduced a fiber-optic distributed acoustic sensing (DAS) method for detecting RPW larvae via trunk vibrations, avoiding electromagnetic interference and scaling effectively for plantation-level monitoring. Belal and Ismail [9] proposed in 2020 a cloud-connected IoT system with Raspberry Pi, microphones, and a mobile app interface for remote pest monitoring and instant alerts. Kavitha et al. [10] in 2021 developed a peak detection and band-pass filtering method tailored for low-power edge devices used in rural agriculture. Abo-El-Saad et al. [11] designed an artificial neural network (ANN) that classified RPW audio spectrograms with over 90% accuracy. In 2022, Al-Hussein et al. [12] utilized convolutional neural networks (CNNs) on mel-spectrograms to address variations in tree species and infestation depth, creating a more universal detection model. Further study by

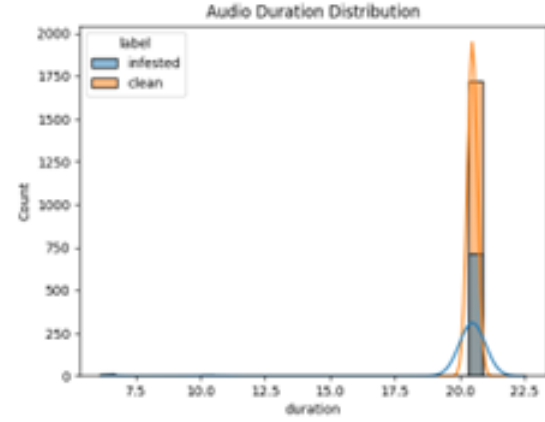


Fig. 1: Distribution plot representing the audio duration distribution for two classes

El-Zoghby et al. [13] demonstrated that sensor placement near the lower trunk, close to feeding zones, significantly enhanced detection accuracy. Rahim et al. [14] in 2023 leveraged transfer learning with the VGGish model for audio event detection, achieving superior results on small datasets. Ahmed et al. [15] introduced a hybrid system fusing fiber-optic sensors with AI-powered processing to enable instant detection and farm-scale deployment.

III. EXPERIMENTAL SETUP

A. Dataset

The data collection named `features_mfcc_contrast.csv` [16] holds 5,153 records with 22 columns to support audio classification work. The dataset stores individual audio samples through the combined information of Mel-Frequency Cepstral Coefficients (MFCC) and spectral contrast features. The dataset includes 13 MFCC features starting from `mfcc_1` to `mfcc_13` that represent sound power distribution across brief time intervals. The dataset contains 7 spectral contrast features identified as `contrast_1` through `contrast_7` that measure the amplitude difference between spectral peaks and valleys. The features serve as discrimination tools for detecting harmonic and percussive elements in audio content and enable precise identification of musical instruments, speech tones and environmental sounds. Fig. 1 to Fig. 4 represents the dataset and preprocessing models. The dataset comprises 731 infested sound files and 1734 non-infested sound files in .wav format. Additionally, 692 test case files were provided in the dataset.

B. Data Preprocessing

The dataset was visually explored to understand how audio duration and volume vary across different classes. Differences in amplitude—essentially the loudness—offered useful clues for identifying whether a tree might be infested. To show these patterns clearly, histograms and box plots were used. Waveforms displayed how the sound levels changed over time, revealing noticeable patterns that helped distinguish

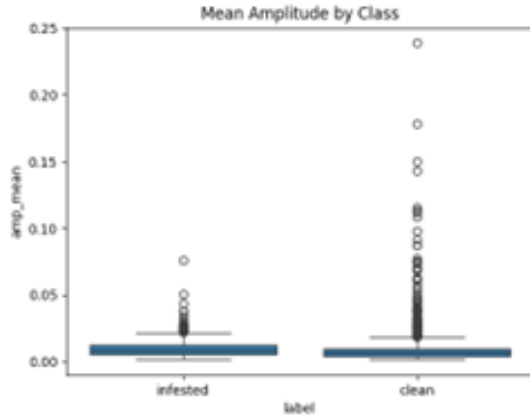


Fig. 2: Mean Amplitude by the class between Infested and Clean

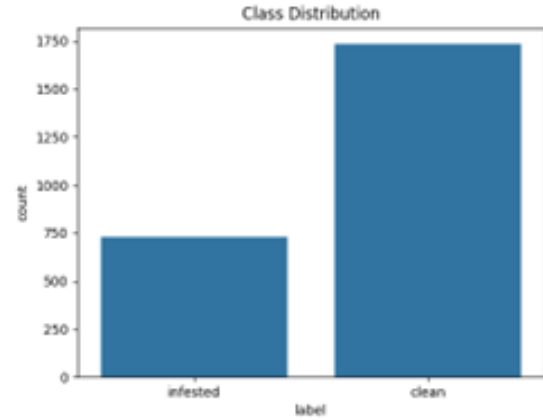


Fig. 4: Distribution of Labels (Infested and Clean)

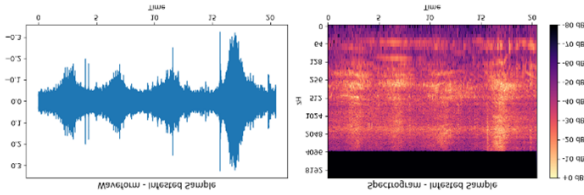


Fig. 3: Waveform and Spectrogram Infested Sample Representation

infested trees from healthy ones. Spectrograms added more detail by showing how the audio frequencies shifted over time, highlighting the non-stationary nature of the signals. For a deeper look, Continuous Wavelet Transform (CWT) graphs were used to capture both time and frequency information. These visuals showed quick, burst-like activity in recordings from infested trees, while the clean ones had smoother, more consistent patterns.

Sound files from each folder were visualized to set noise removal boundaries using bandpass filters. However, varying amplitudes across folders made it difficult to choose optimal frequency limits. Applying a consistent amplitude scale helped achieve uniform results. Two denoising methods were used: Singular Value Decomposition (SVD), which removes low-energy noise components, and bandpass filtering, which isolates useful frequencies between 1kHz and 8kHz. While effective, this combination sometimes removed key signal details. To address this, Wavelet denoising was introduced, preserving important features by filtering noise across different frequency levels. The cleaned files were then visualized and stored.

1) *Normalization*: Normalization of data is a significant step for classification or machine learning processes. Standard normalization is applied for easier feature extraction with MFCC

2) *Segmentation*: This approach segments sound files based on energy levels by calculating RMS energy over short frames.

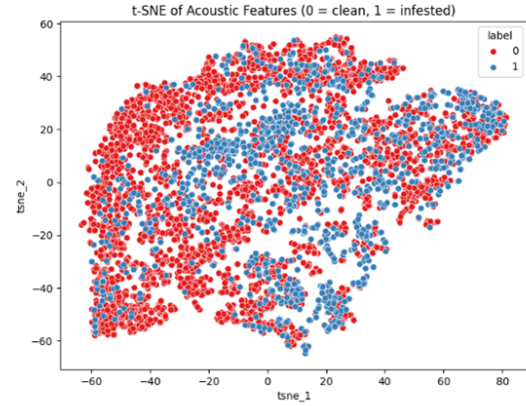


Fig. 5: Feature Visualization from the dataset for the performance

A new segment starts when energy exceeds a set threshold and is kept if it lasts longer than the minimum duration. Clean and infected audio segments are stored separately, speeding up and improving the training process. These segments are then used for feature extraction, helping the model focus on relevant data while excluding noise.

3) *Feature selection* : Features were selected using Recursive Feature Elimination (RFE), which trains a tree-based model to rank and remove the least important features iteratively until no further improvement occurs. The final set of relevant features is saved for training and visualized using t-SNE, a technique that reduces high-dimensional data into lower dimensions for clearer visualization. Fig. 5 shows this feature visualization, which maps data similarities into probabilities to preserve structure between high- and low-dimensional spaces.

4) *Dataset Balancing*: The audio was extremely imbalanced. An oversampling technique was utilized, Synthetic Minority Over-sampling Technique (SMOTE) generates synthetic samples for the minority class by interpolating between ex-

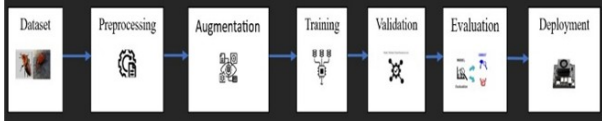


Fig. 6: Data Preprocessing Steps

isting samples, helping to create a more balanced dataset for training machine learning models.

5) *Model selection*: Several models—including XGBoost, LightGBM, CatBoost, AdaBoost, MLPs, Neural Networks, Gradient Boosting, and Histogram Boosting—were trained and evaluated. XGBoost is an optimized gradient boosting method focused on speed and performance. LightGBM uses a histogram-based, leaf-wise tree growth approach for faster training. CatBoost handles categorical data efficiently with minimal preprocessing. AdaBoost combines multiple weak classifiers into a strong one. Combining MLPs with XGBoost and CatBoost leverages both neural networks and boosting to better capture complex data patterns. Neural networks mimic brain functions to learn intricate patterns. Fig. 6 outlines the entire sound detection process before actual detection starts. The proposed model is given detail below.

IV. METHODOLOGY

In the proposed system for early detection of red palm weevil activity using sound, a Feedforward Neural Network (FFNN) is employed as the core classification model. The network processes high-dimensional audio features—specifically Mel Frequency Cepstral Coefficients (MFCC) and spectral contrast values—to determine whether the recorded sound indicates harmful insect activity.

A. Input to the Network

The input to the neural network is a numerical feature vector extracted from each audio file. These features are already scaled to ensure uniform influence across dimensions. Each feature vector serves as a single data point representing the audio signature of a recorded sound.

B. Layer-Wise Flow and Transformation

The FFNN consists of a sequential architecture of fully connected (dense) layers that progressively transform the input feature space into a decision boundary for binary classification:

1) *First Layer (Input $\rightarrow$ Hidden)*: The input feature vector is passed into the first fully connected layer consisting of 512 neurons. Each neuron computes a weighted sum of all input features, adds a bias term, and passes the result through a ReLU (Rectified Linear Unit) activation function. This activation introduces non-linearity, allowing the network to learn complex patterns in the sound data.

2) *Second Layer (Hidden)*: The output of the first layer becomes the input to the second layer with 256 neurons. The same weighted summation and ReLU activation process occurs, enabling deeper abstraction of audio patterns.

3) *Third Layer (Hidden)*: This layer contains 128 neurons and continues the transformation process with ReLU activation. By this stage, the model has learned multiple hierarchical representations of the input features, each highlighting different sound characteristics that are significant for detection.

4) *Final Layer (Output)*: The last layer is a single neuron connected to a Sigmoid activation function, which compresses the output to a probability value between 0 and 1. This value represents the likelihood that the input sound corresponds to harmful red palm weevil activity.

C. Decision Mechanism

After the forward pass through the network, the output probability is interpreted as follows:

- If the probability is greater than 0.5, the model predicts "presence of harmful insect activity."
- If the probability is less than or equal to 0.5, it predicts "no harmful activity." This binary classification threshold allows the system to make interpretable and actionable predictions for pest management.

D. Learning and Optimization

The network learns to make accurate predictions through supervised training using labeled data:

The Binary Cross Entropy Loss measures the difference between predicted probabilities and actual class labels (0 or 1).

The Adam optimizer updates the weights of the network using gradients computed via backpropagation, which adjusts model parameters to reduce the loss function iteratively. This training continues over multiple epochs, with batches of data being processed at each step to improve generalization and convergence.

E. Generalization and Prediction

Once trained, the neural network captures hidden relationships between sound features and pest activity. It generalizes these patterns to new, unseen audio samples, enabling real-time detection with high accuracy. The model is evaluated on a separate test set to ensure its effectiveness outside the training environment.

F. Architecture of model

Fig. 7 represents the structure of the training model.

V. RESULT AND DISCUSSION

The proposed neural network model achieved a testing accuracy of 85.06% and training accuracy of 94.04%, demonstrating strong overall classification performance. The precision of 0.69 indicates a good level of correctness in positive predictions, while the recall of 0.80 reflects the model's effectiveness in identifying true positive cases. The F1-score of 0.74 balances precision and recall, highlighting the model's robustness in handling imbalanced data and ensuring reliable detection of the minority class and support of 269 from the infested output. These results confirm that the neural network architecture,

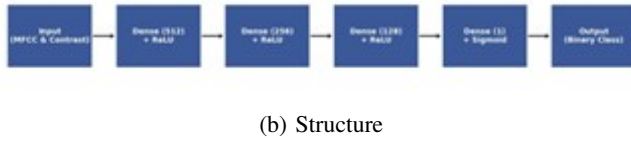
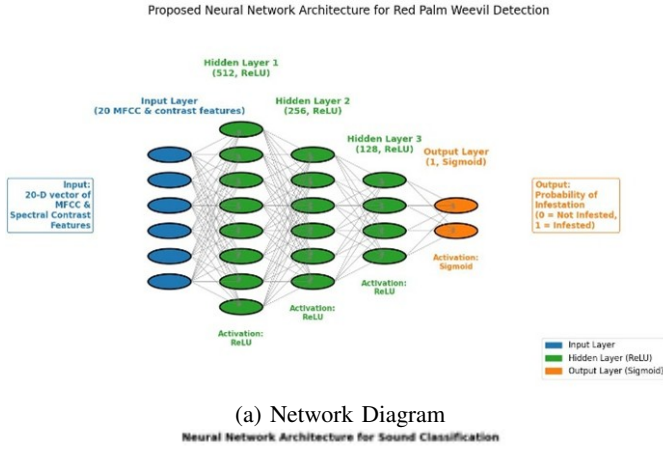


Fig. 7: Architecture Diagram for the Neural Network model

enhanced with SMOTE for class balancing, provides effective discrimination between healthy and infested coconut trees using acoustic features. The high recall value is particularly significant for pest detection applications, where missing an infestation (false negative) can have severe agricultural and economic consequences. Based on the performance of the model the heatmap is generated below.

VI. MODEL PERFORMANCE

To assess the classification performance of the neural network model for detecting infestation in coconut trees, a confusion matrix was generated using the true and predicted labels from the test set. The confusion matrix Figure 8 and 9 provides a detailed breakdown of correct and incorrect predictions for each class, allowing for a clear visualization of model strengths and weaknesses. Below values represents the confusion matrix of the predicted label.(Fig. 8)

- True Negatives (Healthy): 667
- False Positives: 95
- False Negatives: 59
- True Positives (Infested): 210

This heatmap enables intuitive understanding of the model's prediction distribution across both classes(Fig. 9) Performance Summary:

- Accuracy(Testing): 85.06%
- Accuracy (Training): 95.04%
- Precision (Infested): 0.69
- Recall (Infested): 0.80
- F1-score (Infested): 0.74

The confusion matrix shows the model correctly identified 210 infested and 667 healthy coconut tree samples, with 59 infested trees missed (false negatives) and 95 healthy trees wrongly flagged (false positives). The model's high recall for infested

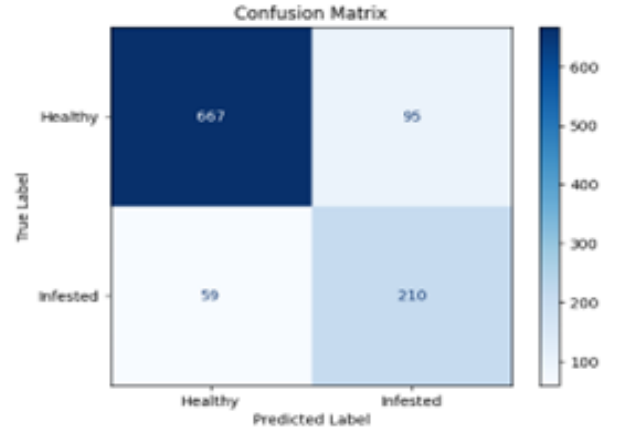


Fig. 8: Model performance between the Healthy and Infested

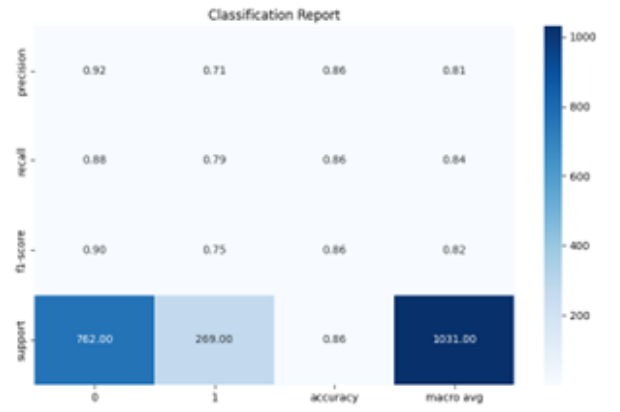


Fig. 9: Heatmap visualization of a classification report, showing precision, recall, F1-score, and support for each class, along with overall accuracy and macro averages.

trees is key, as it reduces missed detections. Overall, the neural network performed well, accurately detecting infestations and balancing sensitivity and specificity. This makes it a reliable tool for early pest detection in agriculture.

VII. EVALUATION METRICS AND STATISTICAL ANALYSIS

To comprehensively assess the performance of the neural network, several standard metrics for binary classification were used. The neural network model's performance for detecting pest infestations in coconut trees was evaluated using precision, recall, and F1-score. The model achieved a precision of 0.69, indicating that 69% of the trees predicted as infested were indeed infested. The recall was 0.78, reflecting the model's ability to correctly identify 78% of all truly infested trees. The F1-score, which balances both precision and recall, was 0.73, demonstrating that the model effectively manages the trade-off between false positives and false negatives. These values suggest that the model is both reliable in identifying infested trees and efficient in minimizing unnecessary false alarms.

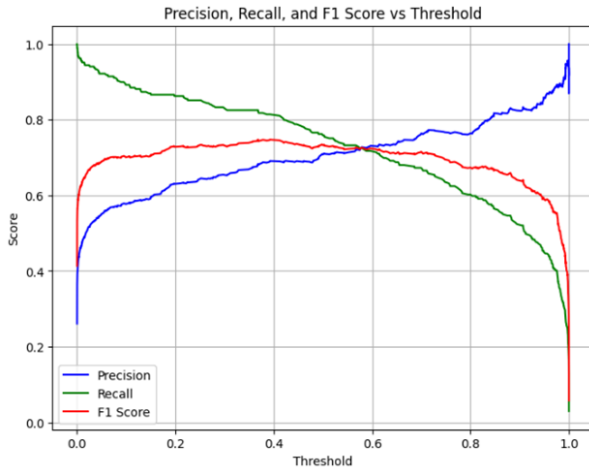


Fig. 10: Metrics change as the classification threshold is varied between the precision, recall, F1-Score

The attached line graph, titled Fig. 10 visually illustrates how these metrics change as the classification threshold is varied. As the threshold increases, precision (blue line) rises while recall (green line) declines, highlighting the typical trade-off between these metrics. The F1-score (red line) peaks at an intermediate threshold, indicating the optimal balance point for the model's predictions. This visualization is particularly useful for selecting the most appropriate threshold for real-world deployment, ensuring that the model can be tuned to prioritize either sensitivity or specificity based on operational requirements. Overall, the statistical analysis and graphical representation confirm the model's robustness and practical value in early pest detection for coconut trees. Now we compare the proposed model with the other model to come to a conclusion.

VIII. COMPARATIVE ANALYSIS

To rigorously evaluate the effectiveness of various machine learning approaches for coconut tree pest detection, we compared a custom PyTorch feedforward neural network (FFNN) with several state-of-the-art ensemble methods, including XGBoost, CatBoost, LightGBM, Gradient Boosting, Histogram-based Gradient Boosting, AdaBoost, a scikit-learn MLP, and a stacking ensemble (MLP + XGBoost + CatBoost). All models were trained and tested on the same dataset, with SMOTE applied to mitigate class imbalance and standardized feature scaling for consistency

A. Outcomes

The PyTorch neural network achieved the highest overall accuracy (85%) and macro-averaged F1-score (0.81), outperforming all other models tested. Its recall (0.75) and F1-score (0.72) for the minority (infested) class were among the best, indicating an excellent balance between sensitivity and specificity—a crucial requirement for real-world pest detection, where both missed detections and false alarms can have significant consequences. The scikit-learn MLP achieved

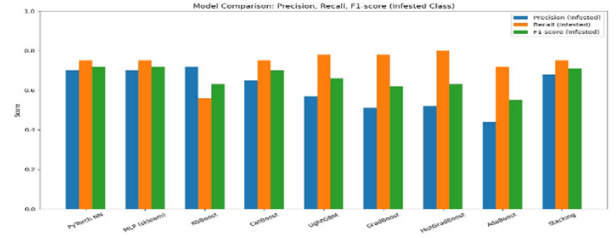


Fig. 11: Comparative analysis based on the precision, F1 score, recall

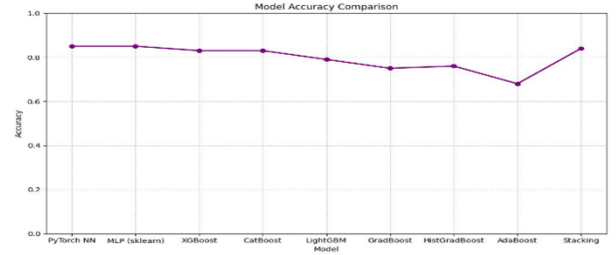


Fig. 12: Comparative analysis based on the Accuracy with different models

nearly identical results, confirming the robustness of neural networks for this application. Among the ensemble models, CatBoost and XGBoost demonstrated strong performance, with CatBoost matching the neural network's recall for the minority class and XGBoost achieving the highest precision among ensembles. However, both fell slightly short of the neural network in terms of overall F1-score and accuracy (Fig. 11). Fig. 12 shows the comparative analysis based on the accuracy with different models. The stacking ensemble, which integrates the predictions of MLP, XGBoost, and CatBoost, also delivered competitive results, closely trailing the neural network in all key metrics. Traditional boosting methods such as AdaBoost and Gradient Boosting underperformed relative to both neural networks and advanced boosting algorithms, particularly in F1-score for the minority class. This finding is consistent with recent literature, which highlights the advantages of deep learning and modern boosting techniques over classical ensemble methods, especially on complex or imbalanced datasets.

IX. CONCLUSION

This study shows that the PyTorch neural network is the most effective model for detecting red palm weevil infestations in coconut trees using sound data. It achieved the highest accuracy (85%) and macro F1-score (0.81), outperforming other advanced models like XGBoost and CatBoost. Its strong recall for infested trees makes it well-suited for early detection, crucial for timely pest management. Using techniques like SMOTE and a carefully designed network, it effectively handles class imbalance and complex acoustic patterns. However, the model requires large labeled datasets, significant computing power, and is less interpretable than

traditional methods. Despite these limitations, it remains the most accurate and reliable solution for automated acoustic pest detection in coconut plantations, supporting better crop protection.

ACKNOWLEDGMENT

The preferred spelling of the word “acknowledgment” in America is without an “e” after the “g”. Avoid the stilted expression “one of us (R. B. G.) thanks ...”. Instead, try “R. B. G. thanks...”. Put sponsor acknowledgments in the unnumbered footnote on the first page.

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